

- 77.** The average of u, v, w, x, y and z is 10. What is the average of $u + 10, v + 20, w + 30, x + 40, y + 50$ and $z + 60$? **2013 II**

(a) 30 (b) 35 (c) 40 (d) 45

- 78.** The mean of the following distribution is 18.

Class interval	Frequency
11-13	3
13-15	6
15-17	9
17-19	13
19-21	f
21-23	5
23-25	4

What is the value of f ? **2014 II**

(a) 8 (b) 9 (c) 10 (d) 11

- 79.** Consider the following statements pertaining to a frequency polygon of a frequency distribution of a continuous variable having seven class intervals of equal width.

- I. The original frequency distribution can be reconstructed from the frequency polygon.
II. The frequency polygon touches the X-axis in its extreme right and extreme left.

Which of the statement(s) given above is/are correct? **2014 II**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 80.** Consider the following in respect of variate which takes values 2, 2, 2, 2, 7, 7, 7 and 7.

- I. The median is equal to mean.
II. The mode is both 2 and 7.

Which of the statement(s) given above is/are correct? **2014 II**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 81.** Which of the following items of information is a good example of statistical data? **2014 II**

- (a) A table of logarithms of numbers
(b) A list of names of 120 students of a class
(c) A list of annual incomes of the members of a club
(d) Holiday list of the offices of Government of India in the year 2013

- 82.** The following table gives 'less than' type frequency distribution of income per day.

Income (in ₹) less than	Number of persons
1500	100
1250	80
1000	70
750	55
500	32
250	12

What is the modal class?

2014 II

- (a) 250-500 (b) 500-750
(c) 750-1000 (d) None of these

- 83.** The geometric mean of $(x_1, x_2, x_3, \dots, x_n)$ is X and the geometric mean of $(y_1, y_2, y_3, \dots, y_n)$ is Y .

Which of the following statement is/are correct?

I. The geometric mean of

$(x_1 y_1, x_2 y_2, x_3 y_3, \dots, x_n y_n)$ is XY .

II. The geometric mean of $\left(\frac{x_1}{y_1}, \frac{x_2}{y_2}, \frac{x_3}{y_3}, \dots, \frac{x_n}{y_n}\right)$ is $\frac{X}{Y}$.

Select the correct answer using the codes given below

- (a) Only I (b) Only II **2014 II**
(c) Both I and II (d) Neither I nor II

- 84.** Consider the following statements in respect of a discrete set of numbers.

- I. The arithmetic mean uses all the data and is always uniquely defined.
II. The median uses only one or two numbers from the data and may not be unique.

Which of the statement(s) given above is/are correct? **2014 II**

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 85.** If x_i 's are the mid-points of the class intervals of grouped data, f_i 's are the corresponding frequencies and $\bar{x}$ is the mean, then what is $\sum f_i(x_i - \bar{x})$ equal to? **2014 II**

- (a) 0 (b) -1 (c) 1 (d) 2

- 86.** When we take class intervals on the X-axis and corresponding frequencies on the Y-axis and draw rectangles with the areas proportional to the frequencies of the respective class intervals, the graph so obtained is called **2014 II**

- (a) bar diagram (b) frequency curve
(c) ogive (d) None of these

- 87.** Consider the following data:

- I. Number of complaints lodged due to road accidents in a state within a year for 5 consecutive years.

- II. Budgetary allocation of the total available funds to the various items of expenditure.

Which of the above data is/are suitable for representation of a pie diagram? **2014 II**

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 88.** Consider the following statements related to cumulative frequency polygon of a frequency distribution, the frequencies being cumulated from the lower end of the range

- I. The cumulative frequency polygon gives an equivalent representation of frequency distribution table.
- II. The cumulative frequency polygon is a closed polygon with one horizontal and one vertical side. The other sides have non-negative slope.

Which of the statement(s) given above is/are correct? **2014 II**

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 89.** The class which has maximum frequency is known as **2014 II**
(a) median class (b) mean class
(c) modal class (d) None of these
- 90.** Ten observations 6, 14, 15, 17, $x + 1$, $2x - 13$, 30, 32, 34 and 43 are written in ascending order. The median of the data is 24. What is the value of x ? **2014 II**
(a) 15 (b) 18 (c) 20 (d) 24

Directions (Q. Nos. 91-94) Consider the following frequency distribution. **2015 I**

Class	Frequency
0-10	4
10-20	5
20-30	7
30-40	10
40-50	12
50-60	8
60-70	4

- 91.** What is the mean of the distribution?
(a) 37.2 (b) 38.1 (c) 39.2 (d) 40.1
- 92.** What is the median class?
(a) 20-30 (b) 30-40
(c) 40-50 (d) 50-60
- 93.** What is the median of the distribution?
(a) 37 (b) 38 (c) 39 (d) 40
- 94.** What is the mode of the distribution?
(a) 38.33 (b) 40.66
(c) 42.66 (d) 43.33
- 95.** There are five parties A, B, C, D and E in an election. Out of total 100000 votes cast, 36000 were cast to party A, 24000 to party B, 18000 to party C, 7000 to party D and rest to party E. What angle will be allocated for party E in the pie chart? **2015 I**
(a) 15° (b) 54° (c) 60° (d) 72°

- 96.** If a variable takes discrete values $a + 4$, $a - 3.5$, $a - 2.5$, $a - 3$, $a - 2$, $a + 0.5$, $a + 5$ and $a - 0.5$, where $a > 0$, then the median of the data set is **2015 II**

(a) $a - 2.5$ (b) $a - 1.25$ (c) $a - 1.5$ (d) $a - 0.75$

- 97.** If each of n numbers $x_i = i$ ($i = 1, 2, 3, \dots, n$) is replaced by $(i + 1)x_i$, then the new mean is **2015 II**

(a) $\frac{n+3}{2}$ (b) $\frac{n(n+1)}{2}$
(c) $\frac{(n+1)(n+2)}{3n}$ (d) $\frac{(n+1)(n+2)}{3}$

- 98.** The weighted arithmetic mean of first 10 natural numbers whose weights are equal to the corresponding numbers is equal to **2015 II**

(a) 7 (b) 14 (c) 35 (d) 38.5

- 99.** The election result in which six parties contested was depicted by a pie chart. Party A had an angle 135° on this pie chart. If it secured 21960 votes, then how many valid votes in total were cast? **2016 I**

(a) 51240 (b) 58560 (c) 78320 (d) 87840

- 100.** The mean and median of 5 observations are 9 and 8, respectively. If 1 is subtracted from each observation, then the new mean and the new median will respectively be **2016 I**

(a) 8 and 7 (b) 9 and 7
(c) 8 and 9 (d) Cannot be determined due to insufficient data

- 101.** The age distribution of 40 children are as follows

Age (in years)	5-6	6-7	7-8	8-9	9-10	10-11
Number of children	4	7	9	12	6	2

Consider the following statements in respect of the above frequency distribution

- I. The median of the age distribution is 7 yr.
II. 70% of the children are in the age group 6-9 yr.
III. The modal age of the children is 8 yr.

Which of the above statements are correct?

2016 I
(a) I and II (b) II and III (c) I and III (d) I, II and III

- 102.** Suppose $x_i = \lambda^i$ for $0 \leq i \leq 10$, where $\lambda > 1$. Which one of the following is correct? **2016 I**

(a) AM < Median (b) GM < Median
(c) GM = Median (d) AM = Median

- 103.** Suppose $x_i = \frac{1}{i}$ for $i = 1, 2, 3, \dots, 11$. Which one of the following is not correct? **2016 I**

(a) AM > $1/6$ (b) GM > $1/6$
(c) HM > $1/6$ (d) Median = HM

ANSWERS

1	c	2	b	3	b	4	c	5	a	6	b	7	c	8	c	9	d	10	b
11	d	12	b	13	d	14	a	15	a	16	c	17	a	18	b	19	b	20	b
21	d	22	a	23	c	24	d	25	a	26	a	27	c	28	b	29	b	30	d
31	c	32	a	33	d	34	b	35	b	36	d	37	c	38	c	39	c	40	b
41	b	42	b	43	b	44	b	45	c	46	b	47	c	48	b	49	c	50	b
51	d	52	d	53	c	54	c	55	b	56	c	57	b	58	d	59	b	60	c
61	b	62	b	63	a	64	b	65	a	66	c	67	b	68	c	69	d	70	b
71	b	72	c	73	a	74	a	75	b	76	a	77	d	78	a	79	a	80	c
81	c	82	b	83	c	84	c	85	a	86	d	87	c	88	a	89	c	90	c
91	a	92	b	93	c	94	d	95	b	96	b	97	d	98	d	99	b	100	a
101	b	102	c	103	d														

HINTS AND SOLUTIONS

- (c) Frequency polygon can be drawn by joining the mid-points of the respective tops in histogram.
- (b) An ogive is used to determine the median.
- (b) Let the lower limit be x . Then, the upper limit of class interval = $x + 10$
 $\therefore$ mid-value = 42
 $\therefore \frac{x + (x + 10)}{2} = 42 \Rightarrow 2x + 10 = 84$
 $\Rightarrow 2x = 74 \Rightarrow x = 37$
 $\therefore$ Lower limit = 37
Upper limit = $37 + 10 = 47$
- (c) Lower class limits are obtained by subtracting 0.5 from the lower limit, so clearly 9.5, 19.5, 29.5 and 39.5 are the actual lower class limits.
- (a) Here, sum of 5 observations
 $= x + (x + 2) + (x + 4) + (x + 6) + (x + 8)$
 $= 5x + 20$
and total number of observation = 5
 $\therefore$ Mean of total observations
 $= \frac{\text{Sum of 5 observation}}{\text{Total number of observation}}$
 $= \frac{5x + 20}{5} = 11 \Rightarrow 5x + 20 = 55$
 $\Rightarrow 5x = 55 - 20 = 35 \Rightarrow x = 7$
 $\therefore$ Mean of first three
 $= \frac{x + (x + 2) + (x + 4)}{3} = \frac{3x + 6}{3}$
 $= \frac{3 \times 7 + 6}{3} = \frac{27}{3} = 9 \quad (\because x = 7)$

Hence, the mean of first three observation is 9.

- (b) Total sum of items
 $= 2 \times 12 + 3 \times 12 + 5 \times 12$
 $= 24 + 36 + 60 = 120$
Total sum of items of first two group
 $= (2 + 3) \times 3 = 15$
Total sum of 5 items of group third
 $= 120 - 15 = 105$
 $\therefore$ Mean of third group = $\frac{105}{5} = 21$
Hence, the mean of third group is 21.
- (c) Given, $n_1 = 7, \bar{x}_1 = 10$
and $n_2 = 3, \bar{x}_2 = 5$
 $\therefore$ Combined mean = $\frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$
 $= \frac{7 \times 10 + 3 \times 5}{7 + 3} = \frac{85}{10} = 8.5$
Hence, the mean of combined set is 8.5.
- (c) Given, total number of students = 50
Total number of students passed
 $= 50 - 10 = 40$
Total marks obtained by failed students
 $= 28 \times 10 = 280$
 $\therefore$ Total marks of 40 students
 $= 2800 - 280 = 2520$
 $\therefore$ Required average of passed students
 $= \frac{2520}{40} = 63$
- (d) Clearly, all the statements are true about median.

- (b) Median is the middle item of the series arranged in ascending or descending order.

- (d) Central angle for 540 marks = 360° .
 $\therefore$ Central angle for 108 marks
 $= \frac{108}{540} \times 360^\circ = 72^\circ$

Here, the student scored 108 marks in Social Science.

- (b) Given, $n_1 = 45, \bar{x}_1 = 2, n_2 = 40$
 $\bar{x}_2 = 2.5$ and $n_3 = 15, \bar{x}_3 = 2$.
 $\therefore$ Required mean
 $= \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2 + n_3 \bar{x}_3}{n_1 + n_2 + n_3}$
 $= \frac{45 \times 2 + 40 \times 2.5 + 15 \times 2}{45 + 40 + 15}$
 $= \frac{90 + 100 + 30}{45 + 40 + 15} = \frac{220}{100} = 2.2$

Hence, the mean of the combined distribution is 2.2.

- (d) Here,
 $\Sigma f_i = (1 + 2 + 3 + \dots + n) = \frac{n(n+1)}{2}$
 $\Sigma f_i \times x_i = \left(1 \times 1 + 2 \times \frac{1}{2} + 3 \times \frac{1}{3} + \dots + n \times \frac{1}{n} \right)$
 $= (1 + 1 + 1 + \dots + n \text{ times}) = n$
 $\therefore \text{Mean} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{n}{\frac{1}{2} n(n+1)} = \frac{2}{n+1}$

14. (a) Geometric mean of 40, 50 and $x = (40 \times 50 \times x)^{1/3}$
 $\Rightarrow (40 \times 50 \times x)^{1/3} = 10$ (given)
 $\Rightarrow 40 \times 50 \times x = 10^3 \Rightarrow x = \frac{1000}{40 \times 50} = \frac{1}{2}$
Hence, the value of x is $\frac{1}{2}$.

15. (a) We know that the sum of deviation from their mean is zero.
 $\therefore x_1 f_1 + x_2 f_2 + \dots + x_n f_n = 0$

16. (c) The height of the bar is not proportional to the frequency of the class.

17. (a)

x	f	$f \times x$
0	4	0
1	f	f
2	9	18
3	g	$3g$
4	4	16
Total	$\Sigma f = 17 + f + g$	$\Sigma fx = 34 + f + 3g$

$$\therefore \text{Mean} = \frac{\Sigma fx}{\Sigma f} = \frac{34 + f + 3g}{17 + f + g} \Rightarrow 2 = \frac{34 + f + 3g}{17 + f + g} \quad (\because \text{mean} = 2)$$

$$\Rightarrow 34 + 2f + 2g = 34 + f + 3g \Rightarrow f - g = 0 \quad \dots(i)$$

$$\text{Given, } \Sigma f = 25$$

$$\therefore 17 + f + g = 25 \Rightarrow f + g = 25 - 17$$

$$\Rightarrow f + g = 8 \quad \dots(ii)$$

$$\text{On adding Eqs. (i) and (ii), we get } 2f = 8 \Rightarrow f = 4$$

$$\text{On putting } f = 4 \text{ Eq. (i), we get } 4 - g = 0 \Rightarrow g = 4$$

18. (b) For district A, as the class 44.0-47.0 has maximum frequency, so it is the modal class.

$$\therefore l = 44, f_1 = 59, f_0 = 36, f_2 = 30, \text{ and } b = 3$$

$$\therefore \text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times b = 44 + \frac{59 - 36}{2 \times 59 - 36 - 30} \times 3$$

$$= 44 + \frac{23 \times 3}{52} = 44 + 1.33 = 45.33$$

For district B, as the class 47.0-50.0 has the maximum frequency, so it is the modal class.

$$\therefore l = 47, f_1 = 54, f_0 = 35, f_2 = 41 \text{ and } b = 3$$

$$\therefore \text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times b = 47 + \frac{54 - 35}{2 \times 54 - 35 - 41} \times 3$$

$$= 47 + \frac{19 \times 3}{32} = 47 + 1.78 = 48.78$$

Thus, the mode for district B is higher than the mode for district A.

19. (b) Given, average score of 50 students = 44

$$\text{Total score} = 44 \times 50 = 2200$$

$$\text{Correct score of 50 students} = (2200 - 73 + 23) = 2150$$

$$\therefore \text{Correct average score} = \frac{2150}{50} = 43$$

20. (b) In the less than type of ogive, the cumulative frequency is plotted against the upper limit of the concerned class interval.

21. (d) Mean of A = $\frac{2+3+7+1+3+2+3}{7} = \frac{21}{7} = 3$

Mode of A = 3, having maximum frequency.

Arrange data in ascending order, we have 1, 2, 2, 3, 3, 3, 7.

and median of A = $\left(\frac{7+1}{2}\right)$ th value = 4th

value = 3

Here, mean, median and mode are equal.

22. (a) Total marks in 3 subjects = $(75 + 80 + 85) = 240$ out of 300.
In any other subject the marks are atleast 0.

$$\text{Average of 4 subjects} = \frac{240}{4} = 60\%$$

$\therefore$ Average cannot be less than 60%.

23. (c) Given, mean = 8.9 and median = 9

$$\therefore \text{Mode} = (3 \times \text{Median}) - (2 \times \text{Mean})$$

$$= (3 \times 9 - 2 \times 8.9) = 27 - 17.8 = 9.2$$

24. (d) If every number of a finite set is increased by any number k , then measure of central tendency should also increase by k .

But geometric mean does not have this property.

25. (a) Given distribution is 1, 3, 5, 7, 9, x , 15, 17, 19, 21.
Number of terms = 10 (even)

$$\therefore \text{Median} = \frac{\text{Value of } \frac{10}{2} \text{th term} + \text{Value of } \left(\frac{10}{2} + 1\right) \text{th term}}{2}$$

$$10 = \frac{\text{Value of 5th term} + \text{Value of 6th term}}{2}$$

$$\Rightarrow 10 = \frac{9 + x}{2} \Rightarrow 20 = 9 + x \Rightarrow x = 11$$

26. (a) Required class boundary = Lower class boundary of lowest class + Width of class $\times$ Number of class
 $= 5.1 + 2.5 \times 10 = 30.1$

27. (c) Arranging the observation in ascending order 2, 3, 4, 6, 7, 7, 8

$\therefore$ Number of items = 7

$$\therefore \text{Median} = \left(\frac{n+1}{2}\right) \text{th value} = \left(\frac{7+1}{2}\right) = 4 \text{th value} = 6$$

So, median = 6 and mode of 2, 9, 3, 4, 9, 6, 9 is 9.

$$\therefore \text{Average of median and mode} = \frac{1}{2}(6 + 9) = 7.5$$

28. (b) Let the mean of observations $x_1, x_2, x_3, \dots, x_n$ be $\bar{x}$.

On subtracting 10 from each, the mean would be $(\bar{x} - 10)$.

$$(\bar{x} - 10) = 60\% \text{ of } \bar{x}$$

$$\Rightarrow (\bar{x} - 10) = \frac{60}{100} \bar{x} = \frac{3}{5} \bar{x} \Rightarrow \left(1 - \frac{3}{5}\right) \bar{x} = 10$$

$$\Rightarrow \frac{2}{5} \bar{x} = 10 \Rightarrow \bar{x} = \frac{5 \times 10}{2} = 25$$

Now, 5 is added to all the given observations, then new mean becomes $= \bar{x} + 5 = 25 + 5 = 30$

29. (b) Range of the data = $120 - 71 = 49$

$$\therefore \text{Class size} = \frac{\text{Range}}{\text{Number of classes}} = \frac{49}{7} = 7$$

The class are 71-78, 78-85, 85-92, 92-97 etc.

So, limits of second class interval are 78 and 85.

30. (d) Here, mean, $M = \frac{y + \frac{1}{y}}{2} \dots(i)$

New mean

$$= \frac{y^3 + \frac{1}{y^3}}{2} = \frac{\left(y + \frac{1}{y}\right)^3 - 3\left(y + \frac{1}{y}\right)}{2}$$

$$= \frac{\left(y + \frac{1}{y}\right) \left[\left(y + \frac{1}{y}\right)^2 - 3\right]}{2} \text{ [from Eq. (i)]}$$

$$= M[(2M)^2 - 3] = M(4M^2 - 3)$$

31. (c) Since, maximum frequency is 80, hence mode will be between 15 to 20.

32. (a) ₹ 25 = 1 cm² $\Rightarrow$ ₹ 1 = $\frac{1}{25}$ cm²

$\therefore$ ₹ 81 = $\frac{81}{25}$ cm² = Area of square

Side of square = $\sqrt{\frac{81}{25}} = \frac{9}{5} = 1.8$ cm

33. (d) Given, $n = 5$

Mean = $\frac{7 + 9 + 11 + 13 + 15}{5} = \frac{55}{5} = 11$

$\Sigma |d_i|^2 = |7 - 11|^2 + |9 - 11|^2 + |11 - 11|^2$
 $+ |13 - 11|^2 + |15 - 11|^2 = 40$

$\therefore \sigma = \sqrt{\frac{\Sigma d_i^2}{n}} = \sqrt{\frac{40}{5}} = 2\sqrt{2} = 2 \times 1.41$
 $= 2.82$

34. (b) Number of cellphones sold by Motorola = $\frac{3}{16} \times 45664$

Central angle of Motorola

$$= \left(\frac{\frac{3}{16} \times 45664}{45664} \times 360^\circ \right) = \left(\frac{3}{16} \times 360^\circ \right)$$

$$= 67.5^\circ$$

35. (b) If the population figures are given for each State of India, then data can be classified as quantitative.

36. (d) The volume of rainfall in certain geographical area, recorded every month for 24 consecutive months.

37. (c) I. First seven prime numbers are 2, 3, 5, 7, 11, 13, 17.

Mean = $\frac{2 + 3 + 5 + 7 + 11 + 13 + 17}{7}$
 $= \frac{58}{7} = 8.28$

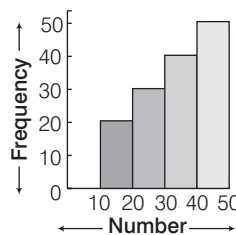
Median = Value of $\left(\frac{7+1}{2}\right)$ th term

= Value of 4th term = 7

Since, prime numbers are always increasing, therefore here mean is always greater than median.

Hence, both statements are correct.

38. (c) Since, the histogram consists of vertical rectangular bars with a common base such that there is no gap between consecutive bars and also the height of the rectangle is determined by the frequency of the class it represents. Hence, both statements are correct in respect of a histogram.



39. (c) We know that, histogram is an equivalent graphical representation of the frequency distribution and is suitable for continuous random variables where the total frequency of an interval is evenly distributed over the interval.

Hence, both the given statements are correct.

40. (b) Primary data is original research that is obtained through first hand investigation. So, statements I and III are primary data.

41. (b) Collar sizes of 200 shirts sold in a week, here mode is a suitable measure of central tendency.

42. (b) Since, the frequency in a straight line, so we take all classes have equal frequency, i.e. 10.

I. It is true that first and last class have 10 frequency.

II. Both the middle classes have frequency, $10 + 10 = 20$

III. Since, all have equal frequency, so we cannot determine the mode.

43. (b) A discrete frequency distribution is such a distribution in which data are presented in a way that exact measurements of the units are clearly shown. Clearly weights of a set of a students is continuous, while other three are discrete.

44. (b) Since, modal size of given data is 1 and its corresponding frequency is 6.

So, x cannot be equal to or more than 6.

Hence, the maximum value of x is 5.

45. (c)

Number of goals	Frequency	CF
0	3	3
1	6	9
2	2	11
3	x	$11 + x$
4	5	$16 + x$
Total	$N = 16 + x$	

Since, median is 2 and the corresponding CF is 11.

So, $\frac{N}{2}$ must be greater than 9.

$\Rightarrow \frac{16 + x}{2} > 9 \Rightarrow x > (18 - 16) \Rightarrow x > 2$

So, smallest possible value of $x = 3$.

46. (b) Here, total expenditure
 $= 6 + 7 + 3 + 4 + 25 + 1.5$
 $= ₹ 24$ lakh

Sectorial angle of largest sector
 $= \frac{360^\circ}{24} \times 7 = 105^\circ$

47. (c) Difference in the sectorial angles of the largest and the smallest sectors

$= 105^\circ - \frac{360^\circ}{24} \times 1.5$
 $= 105^\circ - 22.5^\circ = 82.5^\circ$

48. (b) Here, $l = 18$, $\frac{N}{2} = 50$, $c = 7 + x$,

$f = 40$ and $h = 5$

$\therefore$ Median = $l + \frac{\left(\frac{N}{2} - c\right)}{f} \times b$
 $= 18 + \frac{(50 - (7 + x))}{40} \times 5$

$\therefore 20 = 18 + \frac{5}{40} (43 - x)$

[$\because$ median = 20]

$\Rightarrow 800 = 720 + 215 - 5x$

$\Rightarrow 5x = 935 - 800$

$\Rightarrow 5x = 135 \Rightarrow x = \frac{135}{5} \Rightarrow x = 27$

Hence, the value of x is 27.

49. (c) We have,

$7 + x + 40 + y + 10 + 2 = 100$

$x = 27$

$\Rightarrow 86 + y = 100 \Rightarrow y = 100 - 86$

$\Rightarrow y = 14$

Hence, the value of y is 14.

Solutions (Q. Nos. 50-52) Since, $a < b < c$

Therefore, series is in increasing order i.e. a, a, b, b, c, c

$\therefore$ Median

$$= \frac{\left(\frac{6}{2}\right)\text{th term} + \left(\frac{6}{2} + 1\right)\text{th term}}{2}$$

$$= \frac{3\text{rd term} + 4\text{th term}}{2}$$

$\Rightarrow 2 = \frac{b + b}{2} \Rightarrow b = 2$

Also, arithmetic mean

$$= \frac{a + a + b + b + c + c}{6}$$

$\Rightarrow \frac{7}{3} = \frac{a + b + c}{3} \Rightarrow a + c + 2 = 7$